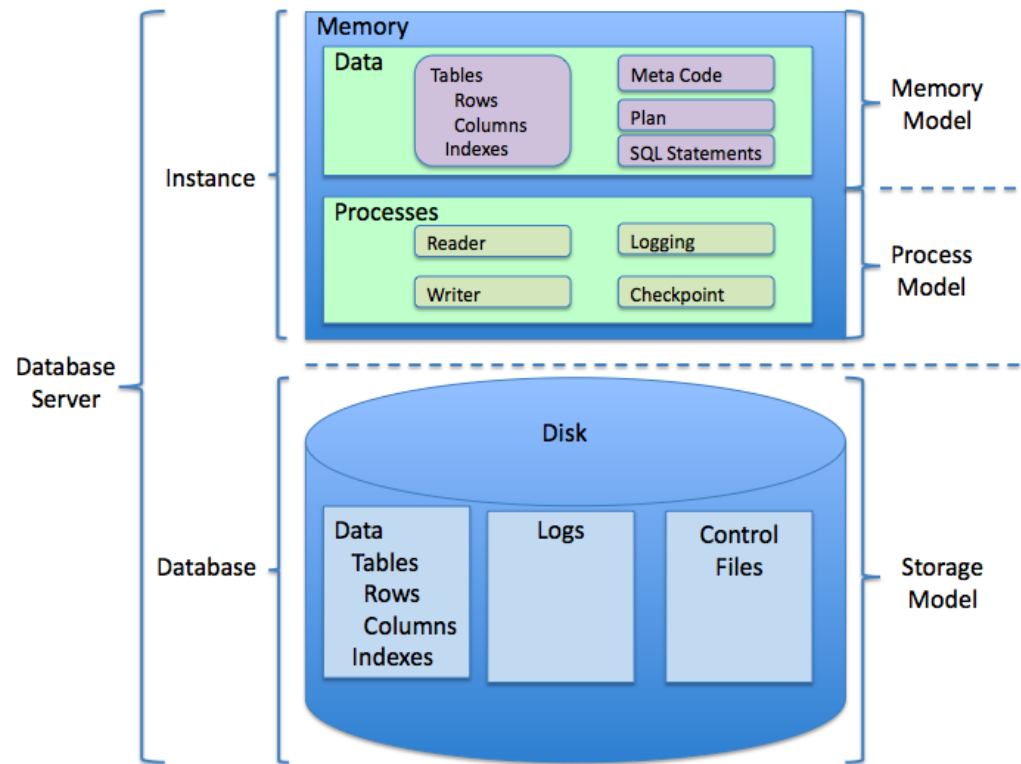


Database Management System



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Types of Keys and concept of Functional Dependency

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Concept of Keys

❖ Key is an attribute or a group of attributes that can determine other attributes. There are several different types of keys

- Super key
- Candidate key
- Primary key
- Surrogate key
- Foreign key
- Secondary key / Alternate Key

Super Key

- ❖ A super key is a key that can uniquely identify any row in the table. In other words super key functionally determines every attribute in the row.
- ❖ **ID, Name, Phone** → Address, Percentage
- ❖ **ID, Percentage** → Name, Phone, Address

Candidate Key

- ❖ It is an attribute or combination of attributes that uniquely identifies a row in a relation. It must satisfy the following properties
 - Unique identification: for every row, the value of the key must uniquely identify that row
 - Non redundancy: No attribute in the key can be deleted without destroying the property of unique identification
- ❖ A minimal (irreducible) superkey. A superkey that does not contain a subset of attributes that is itself a superkey

Primary Key

- ❖ A candidate key selected to uniquely identify all other attribute values in any given row; cannot contain null entries

Foreign Key

- ❖ An attribute or combination of attributes in one table whose values must either match the primary key in another table or be null

Surrogate key

- ❖ An artificial column that is added to a table to serve as the primary key. The DBMS assigns a unique value to a surrogate key when the row is created. Surrogate key is used when primary key consists on multiple columns.
- ❖ For example ***Rental_Property*** (Street, City, Province, PostalCode, Country, Rental_Rate) → ***Rental_Property*** (PropertyID, Street, City, Province, PostalCode, Country, Rental_Rate)

Secondary Key or Alternate Key

- ❖ Field or attribute within a database table that is not the primary key but is still used to efficiently retrieve and access data. Secondary keys are also known as "non-primary keys" or "alternate keys."

Functional Dependency

- ❖ **Determination:** In the context of a database table, the statement “*A determines B*” shows that *if a value of A is known the value of B can be looked up*. In this case A is called *Determinant* and B is called *Dependent*
- ❖ **Determinant:** Any attribute in a specific row whose value directly determines other values in that row e.g. *ID* → *Name*
- ❖ **Dependent:** An attribute whose value is determined by another attribute e.g. *ID* → *Name*
- ❖ **Functional Dependence:** Within a relation R, an attribute B is functionally dependent on an attribute A if and only if *a given value of attribute A determines exactly one value of attribute B*.
 - The relationship “*B is dependent on A*” is equivalent to “*A determines B*” and is written as *A* → *B*.

Identify FDs from the following relation

- ❖ Student (NIC_Num, StdName, StdAddress, HighschoolCode, HighschoolName, HighschoolCity, BoardofEdu, Percentage, Priority)
- ❖ $\text{NIC_Num} \rightarrow \text{StdName}$
- ❖ $\text{NIC_Num} \rightarrow \text{StdAddress}$
- ❖ $\text{NIC_Num} \rightarrow \text{Percentage}$
- ❖ $\text{Percentage} \rightarrow \text{Priority}$
- ❖ $\text{HighSchoolCode} \rightarrow \text{HighschoolName, HighschoolCity}$
- ❖ $\text{HighSchoolCode} \rightarrow \text{BoardofEdu}$

Find Functional Dependencies:

- ❖ Apply(NIC_Num, UniversityName, UniversityProvince, DateOfApplication, ProgramID, ProgramName, StudentName)
- ❖ Business Rules:
 - Date of application for each program in a university is different
 - A student can apply only in one program in a university for his entire life
 - For a multi-campus university UniversityName means Name of the Campus
 - There can be only one campus in a province

Solution

- ❖ ProgramID $\rightarrow$ ProgramName
- ❖ programID $\rightarrow$ UniversityName
- ❖ programID $\rightarrow$ DateOfApplication
- ❖ UniversityName $\rightarrow$ province
- ❖ NIC_Num $\rightarrow$ studentName
- ❖ NIC_Num, ProgID $\rightarrow$ UniversityName